

Forecasting Household Energy Consumption

This project looks at time series forecasting to analyze residential energy use. Using machine learning, we develop models for short-term consumption prediction and detect usage patterns.

Problem Overview & Opportunity

Problem Statement

Smart meters/sensors produce massive datasets daily from household consumption. Most households lack visibility into consumption patterns and usage drivers. Energy dynamics can also vary due to weather, appliances, and seasonal behaviors

Solution

Data science offers predictive insights to help save costs by understanding these complex factors. We can use analytics to understand energy patterns and predict energy consumption

What Can This Solution Achieve?

Households

Save energy bills through behavior change.

Gain visibility on which appliances and conditions drive high usage.

Utilities

Better demand forecasting reduces grid strain significantly.

More effective time-of-use pricing and strategic planning capabilities.

Environmental

Reduce unnecessary electricity consumption across residential sectors.

Support sustainable, low-carbon lifestyles through data-driven insights.

Data Foundation-Sprint 1

Dataset Overview

UCI DataSet

- Measurements of electric power consumption in one household with a one-minute sampling rate over a period of almost 4 years.
- This dataset contains 2M+ measurements gathered in a house located in Sceaux (7km of Paris, France) between December 2006 and November 2010 (47 months).
- Time Series 1-minute resolution data, with 6 features
 - Voltage, Current and Global Active Power usage (Target) used for the HouseHold
 - 3 Sub-metering power features: kitchen, laundry, HVAC systems

Weather DataSet

- Historical hourly weather data from Open-Meteo API for France (2006-2010)
- 34K rows with 1 hour averaged data
- Contains 12 weather related features:
 - Temperature, Dewpoint, Relative humidity
 - surface pressure, cloudcover
 - windspeed, windgust, wind direction
 - shortwave, direct, diffuse, radiation

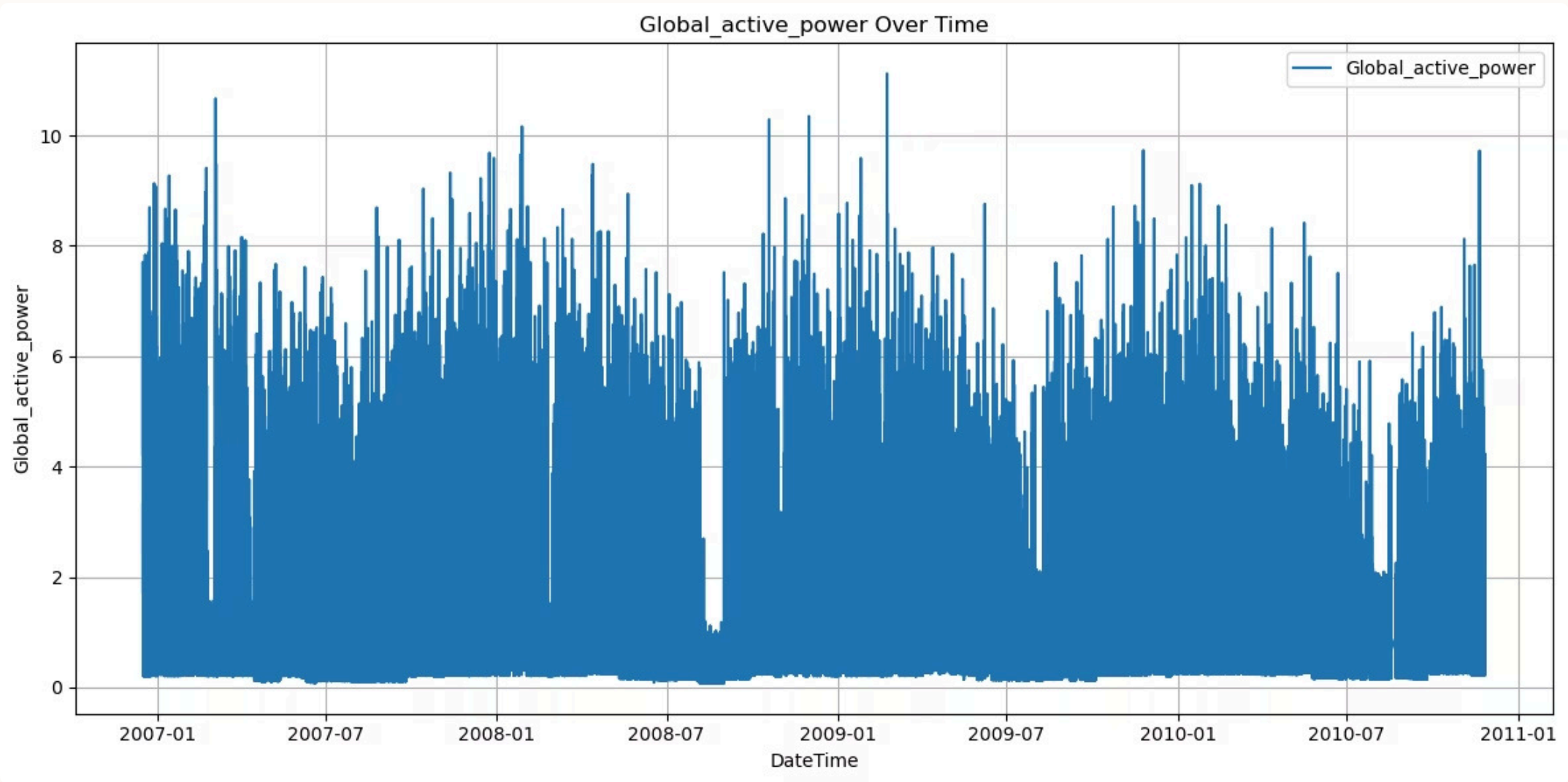
Preprocessing Procedures

Our data preparation involved several critical steps to ensure quality and usability:

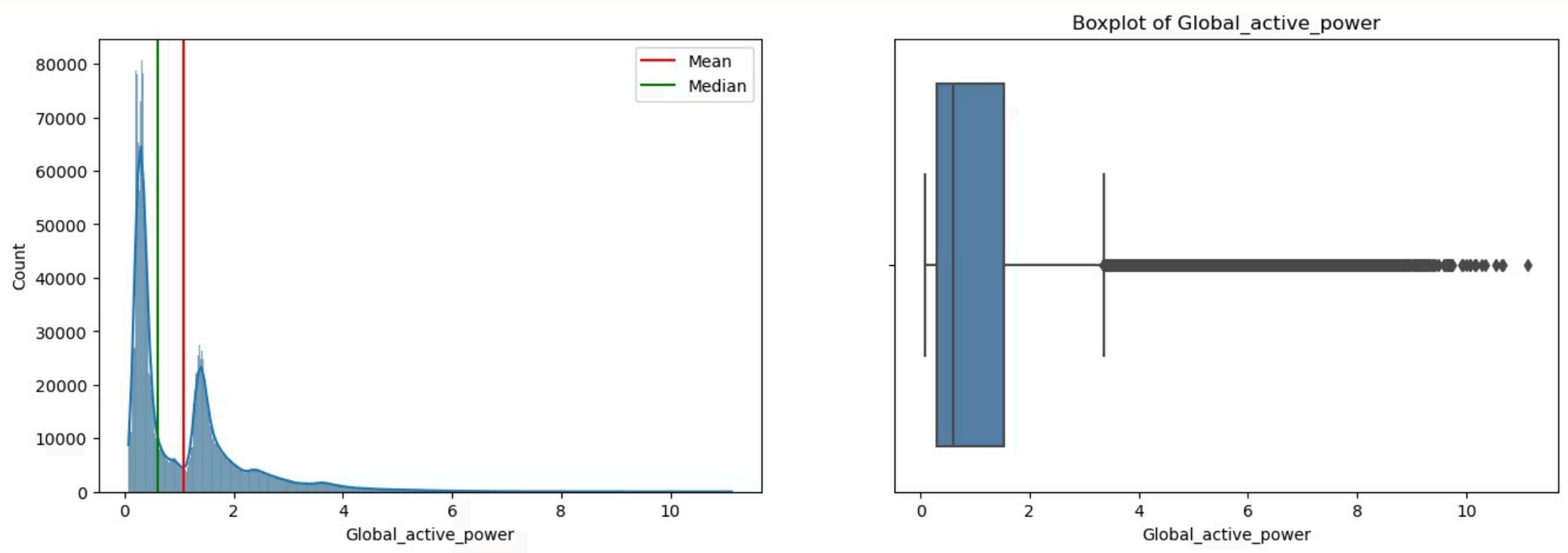
- **Merging Two Datasets:** Integrated both datasets together and resampled the weather data to minute average.
- **Missing Value Treatment:** The UCI dataset contained some missing values in the measurements **~1.25% of the rows**. All calendar timestamps are present in the dataset but for some timestamps, the measurement values were missing. These missing values were from **April 28, 2007**. They were dropped from the dataset, as were not significant compared to the number of rows.
- **Duplicated Data:** Both datasets did not contain any duplicated values.

Key Discoveries from Exploratory Data Analysis -Sprint 1/2

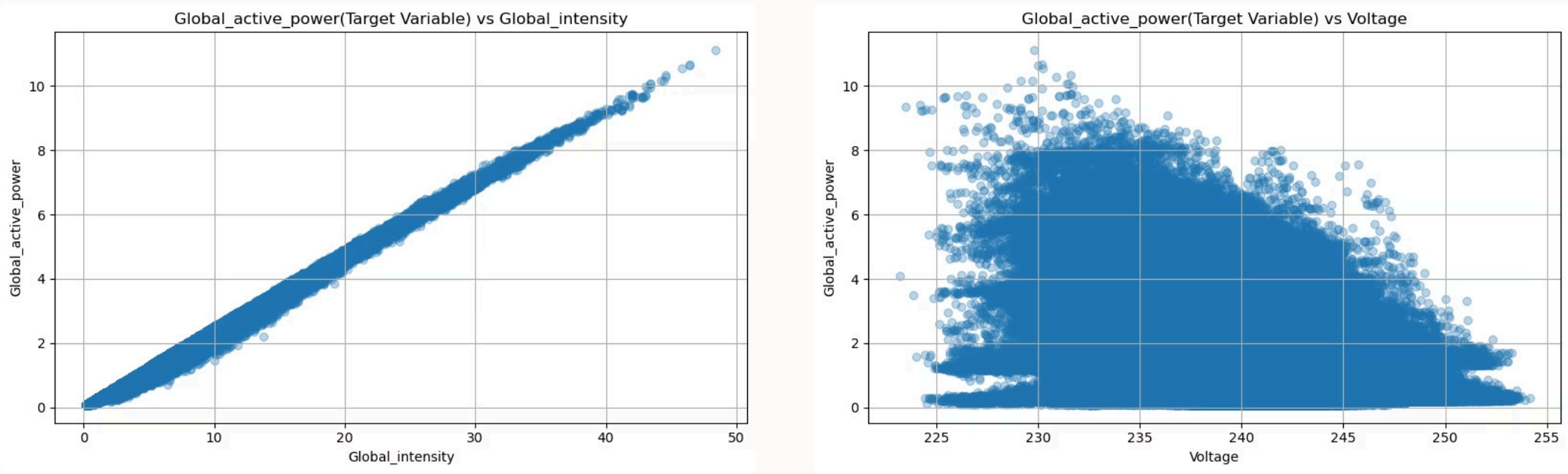
- All features are very noisy as they are sampled every minute, need to consider upsampling the data to **daily average or monthly average**



- The target variable (Power) is **not normally distributed** — it's highly skewed with heavy tails. **Outliers** are frequent; they may represent unusual events or faulty readings. The **mean > median** confirms positive skew.



- High positive correlation between **Target Variable (Power)** and **Intensity (Current)**, this makes sense as **Power= Current * Voltage**, but similar trend was not observed for Voltage



Feature Engineering -Sprint 2

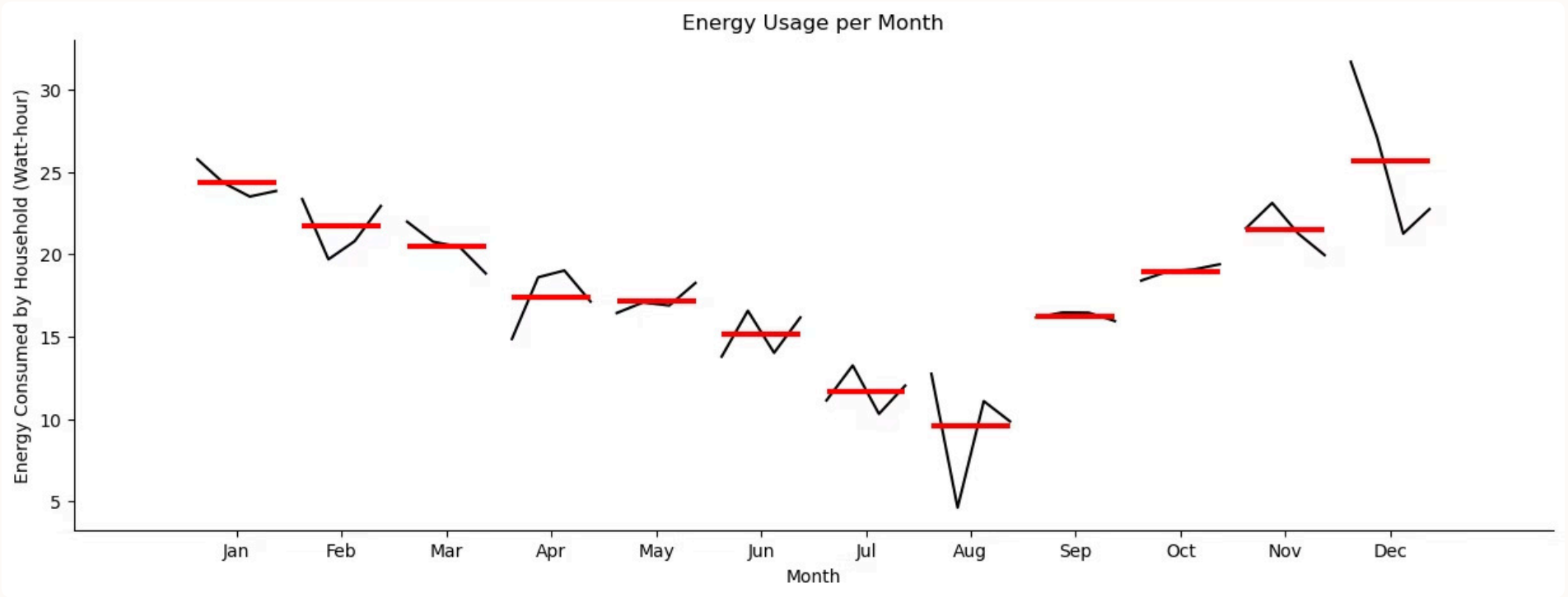
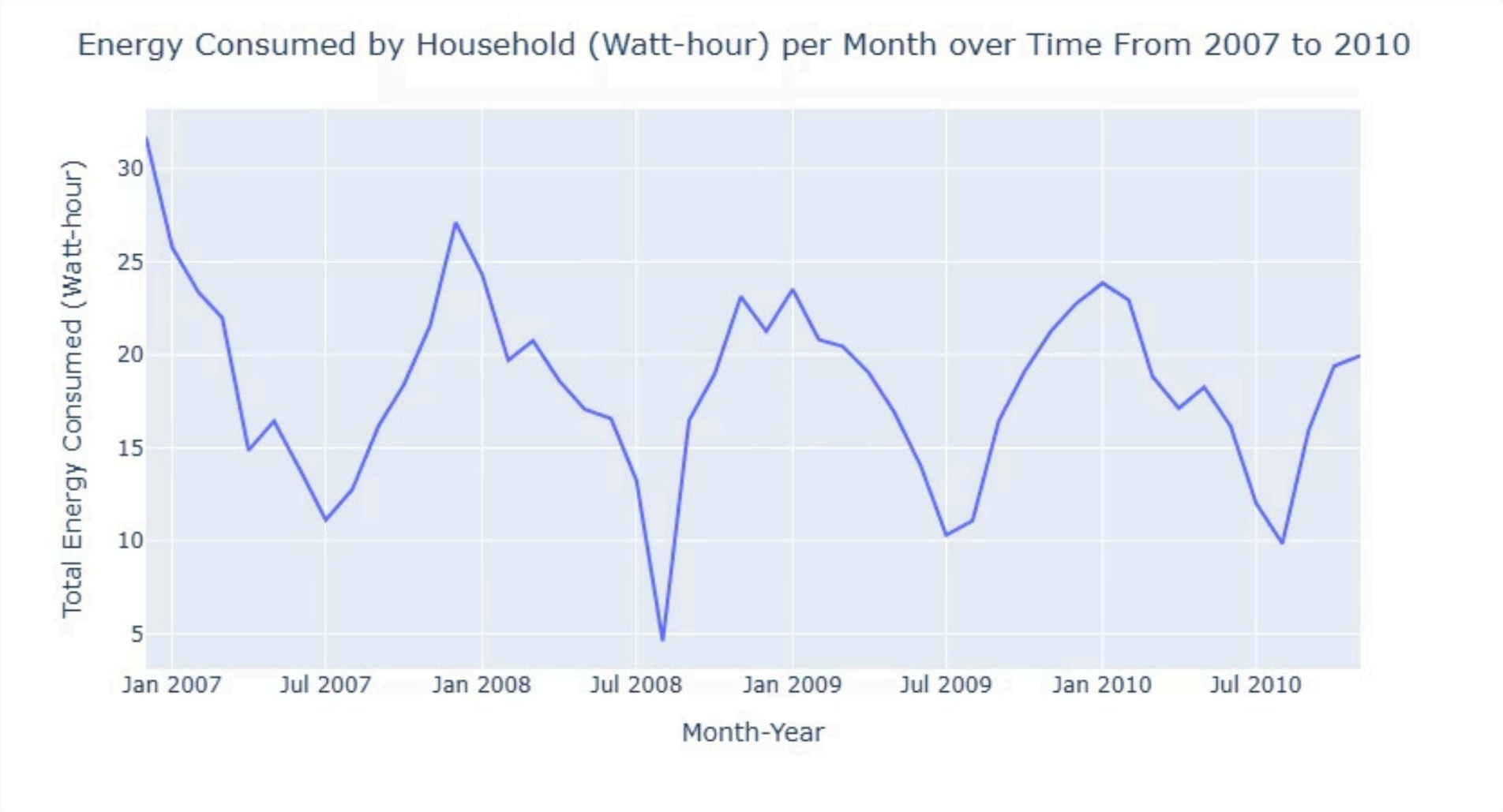
Feature Engineering:

- Created 1 derived feature capturing the **total energy consumed every minute (in watt hour) in the household by electrical equipment not measured in sub-meterings 1, 2 and 3:**
 - **Global Active power*1000/60 - sub_metering_1 - sub_metering_2 - sub_metering_3**
- Performed Unit conversions to change **Power units (kW) to Energy units (kwh)** for features to achieve consistency among submetering 1,2,3 and power variables.
- Resampled the data to **daily, week and monthly** averages to find insights about aily, weekly and sesonal patterns from dataset
- For Linear Regereession model, created a new feature **"Days from Start"**, which resembled the the total days for the 4 years spaced squentially from start month to end month

Advanced EDA Insights-Sprint 2

Seasonal and Monthly Patterns

Identified key monthly and sesonal patterns, where the energy usage is cyclic in nature decreases during summer (July, August) and increases during winter (Jan-Dec).



The red bars shows the average for the given month across all years, while each black line represent an individual month and the progression of values across the years.

Trend Sesonal Decomposition (TSD):



The trend is generally decreasing over time, indicating a long-term decrease in energy consumption by the household. The seasonal component shows a clear pattern of higher usage in winter months and lower usage in summer months, which is consistent with the expectation that households use more energy for heating in winter. The residuals show some random fluctuations around zero, indicating that there are no patterns left after accounting for trend and seasonality.

Baseline Modeling Approach- Sprint 2



Data Splitting

75% training, 25% testing with time-based separation to prevent data leakage



Model Competition

5 candidate algorithms evaluated:
Mean/Average Model, Linear Regression Model with One Feature, Linear Regression Model with One Hot Encoder, Linear Regression Model with multiple dataset features, ARIMA (untuned hyper parameters for now)



Evaluation Metrics

Since we are predicting a continuous numeric variable, time series forecasting is fundamentally a regression problem. Therefore, we need a metric that gives us a sense of the average error of our predictions, so R^2 tends to be less useful in this case. Instead, I have employed the **Mean Absolute Error (MAE)** and the **Mean Absolute Percentage Error (MAPE)** to evaluate and compare our models.

Here is an overview of the models that I will build:

- 1. **Average Model:** $y = mean(y)$
- 2. **Linear Model with One Feature:** $y = \beta_0 + \beta_1 DaysFromStart$
- 3. **Linear Model with Dummy Variables:** $y = \beta_0 + \beta_1 DaysFromStart + \beta_2 January + \beta_3 February + + \beta_{13} December$

4. **Linear Model with Features:** $y = \beta_0 + \beta_1 X1 + \beta_2 X2 + \beta_3 X3 + + \beta_{19} X19$

where, X represents the features we have in the dataset for power df

Baseline Results

Model	Train_MAE	Test_MAE	Train_MAPE	Test_MAPE
Mean?Average	4.034347e+00	3.101882e+00	2.922166e-01	2.158319e-01
linreg	4.077902e+00	3.336767e+00	2.946289e-01	2.017694e-01
linreg_ohe	1.315639e+00	1.717609e+00	9.581909e-02	1.047659e-01
linreg_features	2.021206e-14	1.986290e-14	1.224942e-15	1.145575e-15
arima_311	3.826463e+00	5.577666e+00	2.330022e-01	3.979674e-01

Linear Regression with data set features, is the current best model fitted so far, this would eventually change once I have tuned ARIMA

Next Steps: Advanced Modeling Roadmap-Sprint 3

- Optimize the ARIMA model through Hyper Parameter Tunning
- Look at XGBoost and Facebook's Prophet time series forecasting tool- to complete modelling step
- Look into some Deep Learning models like LSTM, given the time available or for future developments

Thank You